STAFF REVIEW · H446 213 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Insertion sort.

2. Decrease the exponent by the same number of places.

3. ¬A ∧ ¬B.

4. 0.75×8=6.

I Do — source worked steps

1. Predict first — Write +6 as 00000110.

2. Check the method — Invert to 11111001.

3. Check the method — Add one to get 11111010, representing −6.

We Do — guided CHECK

1. Check the method
Invert to 11110110.
Add one to get 11110111, representing −9.

You Do — source mark schemes

1. [8] (a) +19=00010011 [1], invert=11101100 [1], add 1=11101101 [2]. (b) Arithmetic right shift replicates leading 1 [1], gives 11110110 [2], interpreted as −10 [1]; arithmetic right shift of odd negative values rounds towards negative infinity.

2. [8] (a) ¬A ∨ ¬¬B [2], simplifying to ¬A ∨ B [1]. (b) The group covers all eight 3-variable combinations [2]; all three variables vary [1], leaving no literal [1], so output is constant 1 [1].

3. [8] $0.0011\times2^{5} \rightarrow 0.0110\times2^{4}$ [2] $\rightarrow 0.1100\times2^{3}$ [2]; positive normalised mantissa begins 01 [1]; shifting left doubles mantissa and reducing exponent compensates [2]; decimal value 6 [1].

Exit Ticket — source answer

1. False. It produces −10 in two's complement due to sign extension and rounding towards negative infinity.